

Jan 2026 CSE 208

Online on APSP

Duration: 30 minutes

Railway Inspection Route

A railway company manages a transportation network consisting of N stations connected by M one-way railway tracks. The network is represented as a weighted directed graph, where each station corresponds to a vertex and each railway track corresponds to a directed edge. Every edge has a positive integer weight representing the travel time required to traverse that track.

To strengthen security, the company has designated two stations, A and B , as mandatory inspection hubs. Every passenger journey must pass through at least one of these inspection hubs before reaching its destination.

For every ordered pair of stations (source, destination), determine the minimum possible travel time of a valid route from source to destination that satisfies the inspection requirement along with the inspection hub (A or B) through which the optimal route passes.

If no valid route exists, print -1 .

Input

The first line contains two integers N and M — the number of railway stations and the number of one-way railway tracks, respectively ($1 \leq N \leq 500, 1 \leq M \leq N^2$)

Each of the next M lines contains three integers u , v , and t , indicating a directed railway track from station u to station v with a travel time of t ($1 \leq t \leq 10^9$).

The next line contains two integers A and B , representing the two designated inspection hubs.

The following line contains an integer Q , the number of journey queries.

Each of the next Q lines contains two integers s and d, representing the starting station and destination station of a journey.

Output

For each query, print two values:

MinimumTravelTime InspectionHub

where:

MinimumTravelTime is the minimum travel time of a valid journey from station s to station d. InspectionHub is the inspection hub (A or B) through which the optimal route passes.

If both hubs produce the same minimum travel time, print the hub with the smaller station number. If no valid journey exists, print -1.

Example

Sample Input	Sample Output
5 6 0 1 2 1 2 2 2 4 2 0 3 4 3 4 1 1 3 2 1 2 3 0 4 3 2 4 0	5 1 -1 -1

5 9	8 2
1 2 3	11 1
2 1 2	12 1
1 3 4	6 1
3 4 2	6 1
4 1 6	9 1
2 5 3	
5 2 5	
3 2 5	
5 1 9	
1 2	
6	
3 5	
5 3	
4 5	
1 4	
2 3	
4 2	